Lamplight: Manual Functional Test Cases (TC01 to TC28)

Written for: the group, the examiner and anyone repeating the tests on a phone.

These are the device-level tests for the 28 functional cases in the Milestone 03 report (Section 8). Each case lists its steps, expected result and requirement. Record the result, tester and date after each run. Do not mark a case Pass unless you ran it on the installed build.

Environment

- **Build:** release APK (`flutter build apk --release`) or `flutter run` on an Android phone. Record the device model and Android version.
- **Network:** Wi-Fi or 4G for every run. TC28 needs a stated connection.
- **Backend:** cases marked **Live** need the Cloud Functions and rules deployed to the Firebase project. Without them, the case cannot pass.
- **Accounts** (create or confirm before the run):
- `Student A`: `student@lamplight.test`
- `Student B`: a second student account, for the two-user cases (TC11, TC14)
- `Staff`: `staff@lamplight.test`
- Use the passwords you set. Do not write them in this file.
- **Test data:** search for "Introduction to Human Computer Interaction" for TC01 and TC05. Use a book with copies above zero and one with zero for TC03 and TC06. Set copies to zero from the staff Catalog screen if needed.

Result key

- **Pass:** the expected result happened as written.
- **Pass*:** passed after a fix. Note the defect ID (for example D01).
- **Fail:** the expected result did not happen. Log a defect.
- **Blocked:** the case could not run (for example, the backend is not deployed). Say why.

Test cases

ID	Test (operation)	Steps	Expected result	Req.	Live	Result	Tester / date
TC01	Book search: title (Read)	Log in as Student A. Open Books. Type "Introduction to Human Computer Interaction". Submit.	Matching book(s) listed.	FR01	No		
TC02	Search by author, ISBN and subject chip (Read)	Search author "Dix". Clear. Search an ISBN. Clear. Tap a subject chip. Search a nonsense term.	Each search returns matches. The nonsense term shows the empty state.	FR02	No		
TC03	Availability status (Read)	Open a book with copies above zero, then one with zero.	Shows Available and Unavailable correctly.	FR03	No		
TC04	Open book details (Read)	Open any book.	Description, shelf location and due date are shown.	FR04	No		

ID	Test (operation)	Steps	Expected result	Req.	Live	Result	Tester / date
TC05	Reserve an available book (Create)	Open an available book. Tap Reserve. Confirm. Note the copy count before and after.	Reservation saved. Copy count drops by 1.	FR05	Yes		
TC06	Reserve an unavailable book	Open a book with zero copies. Try to reserve.	Reserve is blocked. A waiting list option is offered.	FR05, FR13	Yes		
TC07	Confirmation after reserving (Read)	Complete TC05. Stay on the success screen.	Success screen shows the book, details and pickup deadline.	FR07	No		
TC08	Seat map is live (Read)	Student A opens Seats. Student B books a seat on another phone. Watch Student A's screen without refreshing.	The seat changes to taken on Student A's screen.	FR08	Yes		
TC09	Seat details (Read)	Open Seats. Tap any seat.	Power, quiet and window information is shown.	FR09	No		
TC10	Floor, area and feature filters (Update)	Open the filter sheet. Pick a floor, an area and a feature. Apply.	Map shows only matching seats. A recommended seat is displayed.	FR10	No		
TC11	Two users book the same seat and time	Student A books a seat and time. Student B tries the same seat and time on another phone.	Student B's booking is rejected with a message.	FR11, NFR03	Yes		
TC12	Reserve a seat (Create)	Choose a free seat and time. Confirm.	Reservation saved. Confirmation screen shown.	FR11, FR12	Yes		
TC13	Join the seat waiting list (Create)	Find a full seat. Join the waiting list.	Entry saved. Queue position shown.	FR13	Yes		
TC14	Seat freed while a user is waiting	Student A is on the waiting list for a seat. Student B cancels that seat (or staff releases it).	Student A is notified.	FR14	Yes		
TC15	Seat status after cancel or expiry	Cancel your own seat booking. Check the seat on the map.	Seat shows Available again.	NFR03	Yes		
TC16	Leave the waiting list (Delete)	Open the waiting list entry. Leave it.	Entry removed. Remaining positions update.	FR13	Yes		
TC17	My Reservations (Read)	Open Bookings. Check the Books, Seats and Waiting tabs.	Active book and seat reservations are listed in the right tabs.	FR06	No		
TC18	Cancel a reservation (Update or Delete)	Open a reservation. Tap Cancel. Read the dialog. Confirm.	Dialog states the consequence before confirming. Status becomes cancelled.	FR15	Yes		
TC19	Reservation expires after grace period	Let a reservation pass its pickup deadline and grace period. Wait for the scheduled run.	Status becomes expired. Resource is released. Known issue D02: the expiry can show late until the scheduled run.	FR15	Yes		
TC20	Notifications list (Read)	Open Notifications.	Reminder and status notifications are listed.	FR16, FR18	Yes		

ID	Test (operation)	Steps	Expected result	Req.	Live	Result	Tester / date
TC21	Mark read or dismiss a notification (Update or Delete)	Tap a notification. Mark all read.	Unread marker clears. Item is removed or marked read.	FR18	No		
TC22	Open the QR pass for a seat (Read)	Open a seat reservation. Open the QR pass.	QR code shown with its expiry time.	FR19	Yes		
TC23	Staff scans a valid pass	Staff opens Scan. Scan the QR from TC22.	Result: Valid. Booking status becomes checked in.	FR19	Yes		
TC24	Staff scans an invalid or expired pass	Staff enters a made-up code, or scans an expired pass.	Result: Invalid or Mismatch. No status change.	FR19	Yes		
TC25	Register, log in, wrong password (Create and Read)	Register a new account. Log out. Log in with the wrong password.	Account is created. Wrong password is rejected.	NFR04	Yes		
TC26	Student tries staff screens or another user's data	Log in as Student A. Check there is no staff area. Confirm the rules block reading another user's data (see `rules-tests`).	Staff area is not shown. Other users' data is denied by the rules.	NFR04, NFR05	Yes		
TC27	Contrast, labels and tap targets	Use the Android Accessibility Scanner on key screens (Seats, Books, Bookings). Check status labels appear with colour. Check buttons are at least 48 dp.	Status is never shown by colour alone. Tap targets are at least 48 dp.	NFR06	No		
TC28	Load time of seat map and search	With a stopwatch, open Seats and run a book search, three times each, on the stated connection.	Each load finishes within 3 seconds. Record the three times.	NFR02	Yes		

Notes for the examiner

- **Automated coverage:** the Flutter tests (`flutter test`) and the rules suite (`rules-tests`) cover parts of TC01, TC02, TC09, TC17, TC20, TC21, TC26 and the staff screens. They do not replace the device runs above.
- **Live cases:** when the backend is not deployed, the Live cases are recorded as Blocked, not Pass.
- **Known open defect:** D02 (TC19) is recorded as an open limitation in Section 10 of the report.